AUTOMOBILE ENGINEERING · PART 6 OF 11

# Electric Vehicles

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

## TL;DR

An electric car swaps the engine, fuel tank and gearbox for a battery, an inverter and a motor. The battery's size in **kWh** is its fuel tank; the motor's torque is available from the first turn of the wheels. Running costs per km are far lower than petrol or diesel, but the car costs more to buy, and what matters in practice is whether you can charge at home. Claimed range is a lab figure: the Nexon EV's 489 km claim compares with a user-reported 297 km. GST on EVs is 5%, against 18% or 40% for most petrol and diesel cars.

## 1. How an electric car works

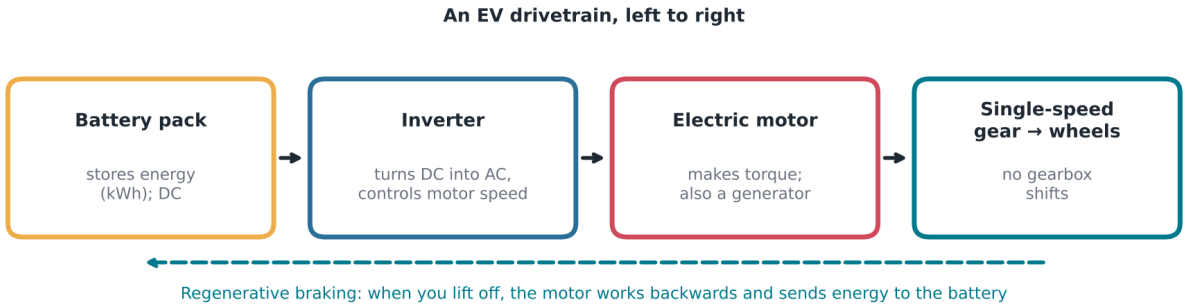



Fig 1. The EV drivetrain: battery, inverter, motor and a single-speed gear. Regeneration sends energy back to the battery.

- **Battery pack:** thousands of cells storing energy as direct current. Capacity is measured in kilowatt-hours (kWh): a 45 kWh pack can deliver 45 kW for one hour.
- **Inverter:** converts the battery's DC into AC for the motor and controls its speed and torque.
- **Motor:** an electric motor makes its peak torque from standstill, with no need to rev up. The Nexon EV (45 kWh) is listed at 145 hp and 215 Nm; the Mahindra BE 6 with the 79 kWh pack at 210 kW (282 hp) and 380 Nm. This is why EVs feel quick off the line.
- **Single-speed gear:** electric motors work over a huge rev range, so one fixed gear is enough. There are no shifts.
- **Regeneration:** lifting off the accelerator turns the motor into a generator, recovering part of the energy that a petrol car loses as brake heat.

## 2. The battery: the costly heart

The battery is the most expensive part of an EV, commonly estimated at 30–40% of the car's cost.* Two chemistries dominate: **LFP** (lithium iron phosphate), cheaper and longer-lived, and **NMC** (nickel-manganese-cobalt), more energy-dense and costlier.* A battery slowly loses capacity over years and charge cycles, which affects resale value; makers typically offer an eight-year battery warranty.* A thermal-management system heats or cools the pack, because batteries age faster when hot.

Importantly, EV platforms put the flat battery in the floor like a skateboard. That lowers the centre of gravity and frees cabin space. The Mahindra BE 6 and XEV 9e sit on Mahindra's INGLO platform; the Tata Punch.ev is the first car on Tata's acti.ev architecture.

## 3. Charging

| Method | Typical power | What it takes (Nexon EV 45 kWh) |
|---|---|---|
| Ordinary 15 A home socket | about 3 kW | about 17.6 hours for 10–100% |
| Home AC wall box | 7.2 kW | about 6.6 hours for 10–100% |
| Public DC fast charger | 60 kW | about 40 minutes for 10–80% |
| Fast charging on newer platforms | 140–180 kW | BE 6: about 20 minutes for 20–80% (59 kWh at 140 kW; 79 kWh at 180 kW) |

A rule of thumb: charging time ≈ energy needed ÷ charger power, plus extra at the top because the car slows charging above about 80%. Fast chargers are best for highway trips; overnight home charging does the daily work. If you cannot charge at home or work, an EV is much less convenient.

## 4. Range: claimed versus real

Claimed range comes from a standard lab test. Real range depends on speed (drag grows with the square, as in Part 2), air conditioning, temperature, load and driving style. The Nexon EV 45 kWh is listed at a claimed 489 km, while one listing gives a user-reported 297 km, about 39% lower. The BE 6 claims 556 km (59 kWh) and about 682 km (79 kWh).* As a quick check, divide the pack size by the range to get consumption: the Nexon's real-world figure works out to about 15 kWh per 100 km, against about 9 kWh per 100 km on the claim.

## 5. The money question

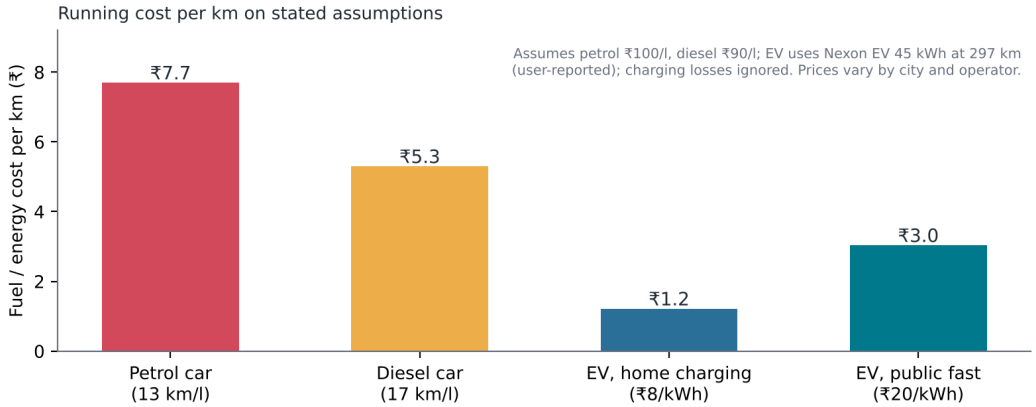



Fig 2. Energy cost per km for a petrol car, a diesel car, and an EV charged at home or at a public fast charger, on the assumptions shown.

On these assumptions, home-charged EV running costs are roughly one-sixth of petrol. Public fast charging is still well under half. But an EV costs more upfront, so the saving only pays back with enough kilometres. A simple break-even: **extra purchase price ÷ saving per km**. If an EV costs ₹4 lakh more and saves ₹6 per km, you need about 67,000 km to recover it (about 5–6 years at 12,000 km a year).

- **Tax:** EVs pay 5% GST; most petrol and diesel cars pay 18% or 40% since 22 September 2025.* This narrows the price gap.
- **Unknowns:** battery life, resale value and charging availability are still uncertain in India.
- **Best fit:** a household with home charging, mostly city and short intercity use, ideally as a second car.

## Summary

- An EV = battery (kWh) + inverter + motor + a single-speed gear; torque is instant.
- The battery is the costly part (roughly 30–40% of the car's cost) and its chemistry and warranty matter.
- Charging time ≈ energy ÷ charger power; home AC does the daily work, DC fast chargers handle trips.
- Claimed range is a lab figure; the Nexon EV example shows real range about 40% below the claim.
- Running cost per km is far below petrol, but the purchase premium means the saving needs enough km to pay back.
- GST is 5% on EVs versus 18% or 40% on most other new cars.

*Unverified or general knowledge: battery cost share (30–40%), LFP vs NMC description, 8-year warranty norm, the BE 6 range figures (sources differ slightly), and the GST statement are not checked against official notices (GST via India Briefing, citing PIB). Specs: Autocar India (Nexon EV: 145 hp, 215 Nm, 489 km claimed, 297 km user-reported; charging times) and Wikipedia (BE 6: 59/79 kWh, 170/210 kW, 380 Nm, 556/682 km, 140/180 kW charging, INGLO). Fuel prices in Fig 2 are assumptions, not quotes. Charging losses (about 10%) are ignored. Break-even example uses made-up figures to show the method.

See also: Part 2 (efficiency) and Part 4 (regenerative braking). Next: Part 7, Hybrids, CNG and ethanol.